

COURSE 3 - DATA ENGINEERING 2026**C3-101 - RetailPulse Capstone Practice Pack**

One progressive repository. Every checkpoint changes the same system and keeps previous evidence.

Artifact	C3-101
Status	2026 - LEARNER EDITION
Phase	C3-09 Production Capstone & Engineering Transition 15 lessons 36-48 focused hours
Prerequisite	C3-08 Gate PASS; all earlier C3 evidence retained.
Assessment boundary	C3-104 is the phase Gate only. C3-113 remains the later course-wide Final Production Gate.
Brand	Charcoal #1F2937 Teal #14B8A6 Soft Blue #3B82F6 AMOLED #000

House teaching rule: requirement -> mental picture -> small proof -> guided integrated build -> expected result -> why it worked -> break/fix -> changed independent case -> validation -> production reasoning -> evidence. Integration is not mastery until the learner can explain and recover the system independently.

Attempt-first workflow

Extract C3_101_RETAILPULSE_CAPSTONE_STARTER_RC1.zip once. Work in that project for P01-P15. Keep the C3-102 review closed until each matching attempt is saved.

Recommended test command: ``python -m pytest -q``. This invocation is portable across the learner venv and avoids depending on how a standalone pytest executable modifies import paths. The starter also includes `pytest.ini` so ``pytest -q`` works in a normal environment.

Evidence convention: `evidence/Lxx/attempt_001/` for first proof, `evidence/Lxx/retry_001/` for a changed retry after review. Never erase failed attempts.

P01 - Capstone ADR: business problem, architecture and non-functional requirements

Your task

Complete docs/ADR_001_RETAILPULSE_TODO.md. Save your actual artifact plus command/run/test/reconciliation or design evidence.

Before you run/change anything

Write the requirement, current input grain/resource boundary, intended output/state, and one invariant that must remain true.

Independent transfer

Change the requirement from daily to sub-minute fraud detection and state exactly which architectural components must change.

Validation to save

ADR names users/data products, measurable NFRs, at least two options, chosen design, rejected alternative and consequences. Tool names are secondary to requirements.

Evidence folder

```
evidence/L01/attempt_001/  
  artifact_or_code  
  observed_result.txt  
  independent_validation.txt  
  failure_recovery_note.md  
  teach_back.txt
```

Only after this folder contains a genuine attempt may you open R01 in C3-102.

P02 - Source contracts, synthetic/public data and reproducible inputs

Your task

Complete configs/source_contracts_TODO.json. Save your actual artifact plus command/run/test/reconciliation or design evidence.

Before you run/change anything

Write the requirement, current input grain/resource boundary, intended output/state, and one invariant that must remain true.

Independent transfer

Add a source field that is optional today but required by a future consumer; define migration and compatibility behavior.

Validation to save

Contracts state explicit grains/keys/change semantics/ownership and distinguish compatible versus breaking evolution. No employer/private data is needed.

Evidence folder

```
evidence/L02/attempt_001/  
  artifact_or_code  
  observed_result.txt  
  independent_validation.txt  
  failure_recovery_note.md  
  teach_back.txt
```

Only after this folder contains a genuine attempt may you open R02 in C3-102.

P03 - Build the batch ingestion path

Your task

Complete scripts/03_batch_ingestion_TODO.py. Save your actual artifact plus command/run/test/reconciliation or design evidence.

Before you run/change anything

Write the requirement, current input grain/resource boundary, intended output/state, and one invariant that must remain true.

Independent transfer

Rerun the same base file twice, then rename it without changing bytes; explain identity behavior.

Validation to save

Local run reports status success, 7 current rows, completed revenue 6910.0 and reconciled=true; source-preserving landing metadata contains hashes/run identity.

Evidence folder

```
evidence/L03/attempt_001/  
  artifact_or_code  
  observed_result.txt  
  independent_validation.txt  
  failure_recovery_note.md  
  teach_back.txt
```

Only after this folder contains a genuine attempt may you open R03 in C3-102.

P04 - Add incremental / CDC-style change handling

Your task

Complete scripts/04_incremental_TODO.py. Save your actual artifact plus command/run/test/reconciliation or design evidence.

Before you run/change anything

Write the requirement, current input grain/resource boundary, intended output/state, and one invariant that must remain true.

Independent transfer

Apply a late correction whose updated_at predates the last observed watermark and explain how your source contract prevents or handles loss.

Validation to save

O102 is 1050.00, O106 and O107 each appear once, current row count is 7, completed revenue remains 6910.0 after replay.

Evidence folder

```
evidence/L04/attempt_001/  
  artifact_or_code  
  observed_result.txt  
  independent_validation.txt  
  failure_recovery_note.md  
  teach_back.txt
```

Only after this folder contains a genuine attempt may you open R04 in C3-102.

P05 - Design layered lakehouse and serving architecture

Your task

Complete architecture/L05_layered_architecture_TODO.md. Save your actual artifact plus command/run/test/reconciliation or design evidence.

Before you run/change anything

Write the requirement, current input grain/resource boundary, intended output/state, and one invariant that must remain true.

Independent transfer

Redesign for a second consumer that needs raw event history instead of current-state orders.

Validation to save

Every layer has a different responsibility/trust contract; serving choice is tied to a real consumer pattern; no layer exists only for naming convention.

Evidence folder

```
evidence/L05/attempt_001/  
  artifact_or_code  
  observed_result.txt  
  independent_validation.txt  
  failure_recovery_note.md  
  teach_back.txt
```

Only after this folder contains a genuine attempt may you open R05 in C3-102.

P06 - Transformation engineering, modeling and tests

Your task

Complete transforms/L06_TRANSFORMATION_AND_DBT_TODO.md. Save your actual artifact plus command/run/test/reconciliation or design evidence.

Before you run/change anything

Write the requirement, current input grain/resource boundary, intended output/state, and one invariant that must remain true.

Independent transfer

Change one business rule for completed revenue and prove the expected result plus regression boundary.

Validation to save

Tests run from a clean project with `python -m pytest -q`; current-state grain and daily sales are independently validated; changed-rule impact is bounded and explained.

Evidence folder

```
evidence/L06/attempt_001/  
  artifact_or_code  
  observed_result.txt  
  independent_validation.txt  
  failure_recovery_note.md  
  teach_back.txt
```

Only after this folder contains a genuine attempt may you open R06 in C3-102.

P07 - Orchestration, retries, backfills and partial reruns

Your task

Complete orchestration/L07_AIRFLOW_DAG_TODO.py. Save your actual artifact plus command/run/test/reconciliation or design evidence.

Before you run/change anything

Write the requirement, current input grain/resource boundary, intended output/state, and one invariant that must remain true.

Independent transfer

Backfill one historical service date after changing only its source file and prove unrelated dates are unchanged.

Validation to save

DAG/design distinguishes retryable vs deterministic failure, scopes work by data interval, supports one bounded backfill and proves partial rerun does not duplicate/corrupt state.

Evidence folder

```
evidence/L07/attempt_001/  
  artifact_or_code  
  observed_result.txt  
  independent_validation.txt  
  failure_recovery_note.md  
  teach_back.txt
```

Only after this folder contains a genuine attempt may you open R07 in C3-102.

P08 - Integrate a real Spark component

Your task

Complete `spark/L08_SPARK_JOB_TODO.py`. Save your actual artifact plus `command/run/test/reconciliation` or design evidence.

Before you run/change anything

Write the requirement, current input grain/resource boundary, intended output/state, and one invariant that must remain true.

Independent transfer

Explain the first bottlenecks/partition changes at 10x and 100x volume without inventing a cluster benchmark.

Validation to save

Spark output matches the local reference on agreed keys/counts/values; scale note explains why Spark is not the cheapest choice for the tiny fixture but what changes at larger scale.

Evidence folder

```
evidence/L08/attempt_001/  
  artifact_or_code  
  observed_result.txt  
  independent_validation.txt  
  failure_recovery_note.md  
  teach_back.txt
```

Only after this folder contains a genuine attempt may you open R08 in C3-102.

P09 - Quality gates and source-to-output reconciliation

Your task

Complete quality/L09_QUALITY_RECONCILIATION_TODO.md. Save your actual artifact plus command/run/test/reconciliation or design evidence.

Before you run/change anything

Write the requirement, current input grain/resource boundary, intended output/state, and one invariant that must remain true.

Independent transfer

Create an offsetting defect that preserves the grand total and prove the key/value reconciliation still catches it.

Validation to save

Quality gate blocks injected corruption even when row count alone could look correct; post-repair keys/values/totals reconcile exactly.

Evidence folder

```
evidence/L09/attempt_001/  
  artifact_or_code  
  observed_result.txt  
  independent_validation.txt  
  failure_recovery_note.md  
  teach_back.txt
```

Only after this folder contains a genuine attempt may you open R09 in C3-102.

P10 - Observability, SLOs and operational monitoring

Your task

Complete observability/L10_SLO_ALERT_TODO.md. Save your actual artifact plus command/run/test/reconciliation or design evidence.

Before you run/change anything

Write the requirement, current input grain/resource boundary, intended output/state, and one invariant that must remain true.

Independent transfer

Change the business cutoff/SLO window and explain the alert effect without changing pipeline code.

Validation to save

SLIs/SLOs are measurable, run evidence is correlated, alerts have owner/action/runbook, and page-worthy versus ticket-worthy conditions are justified.

Evidence folder

```
evidence/L10/attempt_001/  
  artifact_or_code  
  observed_result.txt  
  independent_validation.txt  
  failure_recovery_note.md  
  teach_back.txt
```

Only after this folder contains a genuine attempt may you open R10 in C3-102.

P11 - Security, governance and privacy-safe evidence

Your task

Complete security/L11_ACCESS_GOVERNANCE_TODO.md. Save your actual artifact plus command/run/test/reconciliation or design evidence.

Before you run/change anything

Write the requirement, current input grain/resource boundary, intended output/state, and one invariant that must remain true.

Independent transfer

Add a contractor who can read curated output but must not read raw/landing data; show the control-plane/data-plane distinction.

Validation to save

Static secret scan is clean; permissions are least-privilege by role; one deny boundary is specified/proven where runtime permits; portfolio evidence contains no private secrets/PII.

Evidence folder

```
evidence/L11/attempt_001/  
  artifact_or_code  
  observed_result.txt  
  independent_validation.txt  
  failure_recovery_note.md  
  teach_back.txt
```

Only after this folder contains a genuine attempt may you open R11 in C3-102.

P12 - CI/CD, deployment and controlled change

Your task

Complete `cicd/L12_RELEASE_PIPELINE_TODO.md`. Save your actual artifact plus `command/run/test/reconciliation` or design evidence.

Before you run/change anything

Write the requirement, current input grain/resource boundary, intended output/state, and one invariant that must remain true.

Independent transfer

Introduce a schema/business-rule change and specify rollback versus roll-forward plus post-change reconciliation.

Validation to save

CI fails closed on tests/checks; deploy is protected; credentials are not unnecessarily long-lived; post-deploy smoke includes data reconciliation, not only workflow success.

Evidence folder

```
evidence/L12/attempt_001/  
  artifact_or_code  
  observed_result.txt  
  independent_validation.txt  
  failure_recovery_note.md  
  teach_back.txt
```

Only after this folder contains a genuine attempt may you open R12 in C3-102.

P13 - Incident injection, recovery and prevention

Your task

Complete incident/L13_INCIDENT_REPORT_TODO.md. Save your actual artifact plus command/run/test/reconciliation or design evidence.

Before you run/change anything

Write the requirement, current input grain/resource boundary, intended output/state, and one invariant that must remain true.

Independent transfer

Repeat with a different defect class such as missing customer FK or duplicate order key.

Validation to save

Incident evidence shows defect detection -> containment -> root cause -> bounded repair/replay -> post-recovery reconciliation -> prevention update.

Evidence folder

```
evidence/L13/attempt_001/  
  artifact_or_code  
  observed_result.txt  
  independent_validation.txt  
  failure_recovery_note.md  
  teach_back.txt
```

Only after this folder contains a genuine attempt may you open R13 in C3-102.

P14 - Runbook, README, handoff and portfolio evidence

Your task

Complete docs/L14_HANDOFF_PORTFOLIO_TODO.md. Save your actual artifact plus command/run/test/reconciliation or design evidence.

Before you run/change anything

Write the requirement, current input grain/resource boundary, intended output/state, and one invariant that must remain true.

Independent transfer

Hand the repository to a fresh reviewer using only README/runbook and record every missing instruction.

Validation to save

A fresh reviewer can set up, run, test, troubleshoot and understand evidence using only README/runbook; claims are linked to sanitized proof and limitations are explicit.

Evidence folder

```
evidence/L14/attempt_001/  
  artifact_or_code  
  observed_result.txt  
  independent_validation.txt  
  failure_recovery_note.md  
  teach_back.txt
```

Only after this folder contains a genuine attempt may you open R14 in C3-102.

P15 - Final architecture defense and engineering transition

Your task

Complete docs/L15_ARCHITECTURE_DEFENSE_TODO.md. Save your actual artifact plus command/run/test/reconciliation or design evidence.

Before you run/change anything

Write the requirement, current input grain/resource boundary, intended output/state, and one invariant that must remain true.

Independent transfer

Defend a variant where Fabric is replaced by another lakehouse/cloud platform and separate permanent engineering principles from vendor features.

Validation to save

Defense explains every major component, failure/recovery/evidence path, vendor-specific boundary and 10x/100x change. C3-104 is understood as phase Gate, not graduation.

Evidence folder

```
evidence/L15/attempt_001/  
  artifact_or_code  
  observed_result.txt  
  independent_validation.txt  
  failure_recovery_note.md  
  teach_back.txt
```

Only after this folder contains a genuine attempt may you open R15 in C3-102.